Adaptive Budget Scheduling for Constrained RL Deployment via Energy Composition

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Abstract

Deploying RL policies under cumulative operational constraints (maintenance budgets, duty-cycle limits) typically requires retraining for each constraint specification. We propose a closed-form penalty weight based on the *urgency ratio*—remaining budget divided by remaining time—that composes post-hoc with pre-trained policies via energy addition or gradient correction, requiring zero retraining. A single parameter η controls risk aversion; a single policy serves arbitrary budget levels instantly. On wind farm yaw control, the schedule outperforms an oracle-tuned constant penalty (98.9% vs. 96.7% power retention) and a nonlinear fatigue surrogate achieves 6.8× structural fatigue reduction. We identify the *action-to-cost coupling strength* as the scope condition: post-hoc correction works for action-dependent costs (wind farm) but partially fails for state-dependent costs (Safety Gym), where a learned cost critic Q_c reduces cost by 20–36% but cannot satisfy tight budgets.

1 Introduction

Consider a wind farm operator who trains a reinforcement learning (RL) controller to maximize power production. Six months after deployment, a bearing inspection reveals that one turbine needs restricted yaw travel: it may operate at negative yaw angles for at most 15 hours per year to limit structural fatigue. Retraining the controller is expensive and disruptive. The operator needs a way to impose this new cumulative constraint on the *existing* policy, instantly and without retraining.

This scenario is ubiquitous: cumulative operational constraints arise whenever an agent must limit total time in a costly regime—wind turbines with load budgets [van Dijk et al., 2025], robots with joint-wear cycles, grid batteries with cycle-life limits, and advertisers with spending caps [Balseiro et al., 2024]. Unlike instantaneous constraints, these are *lifetime budgets*: the agent must decide *when* to spend its limited capacity for risky actions.

Inspired by optimal trade execution [Almgren and Chriss, 2001], where a trader paces liquidation based on remaining inventory vs. remaining time, we propose an *urgency ratio* that adapts constraint strength to budget state.

Existing approaches do not solve this: **CMDP methods** [Altman, 1999, Stooke et al., 2020, Sootla et al., 2022] require retraining for each budget specification. **Post-hoc composition** [Urain et al., 2023, Zhang et al., 2025, Xiao et al., 2025] handles only instantaneous constraints, not cumulative budgets spanning an episode.

We fill this gap with three contributions: (1) a **closed-form penalty schedule** $w(u) = \exp(\eta(1/u - 1))$ derived from budget-balanced allocation, parameterized by a single risk-aversion scalar η; (2) **post-hoc composition** with pre-trained policies via energy addition (EBT actors) or gradient correction (MLP actors), enabling zero-retraining budget enforcement; (3) **empirical validation** on wind farm yaw control (6.8× fatigue reduction, 98.9% power) and Safety Gymnasium (diagnostic of state-dependent cost limitations).

2 Background

2.1 Constrained MDPs and Lagrangian Relaxation

A CMDP [Altman, 1999] augments the MDP objective with cost constraints: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ s.t. $\mathbb{E}_\pi[\sum_t c_t^i] \leq d^i$ for each agent i, where $c_t = c(s_t, a_t)$ is the per-step cost and d is the cost limit (following standard CMDP notation [**??**]). The standard Lagrangian approach learns a dual variable μ via primal-dual optimization [Stooke et al., 2020], requiring retraining whenever d changes.

2.2 Energy-Based RL and Post-Hoc Composition

Energy-Based Transformer (EBT) actors [Gladstone et al., 2025] generate actions by gradient descent on a learned energy $E_\phi^i(\mathbf{e}^i, a^i)$. External energy terms compose per-agent before aggregation:

$$E_{\text{total}}(a) = \frac{1}{N} \sum_{i=1}^{N} \left[E_\phi^i(\mathbf{e}^i, a^i) + \lambda_{\text{gs}} \cdot E_{\text{constraint}}^i(a^i) \right] \tag{1}$$

enabling post-hoc constraints without modifying E_ϕ [Urain et al., 2023, Zhang et al., 2025]. We extend this from instantaneous constraints to *cumulative budget constraints* via a time-varying energy weight.

3 Optimal Budget Scheduling

An agent allocates cost limit d over T steps. Per-step cost $c_t := c(s_t, a_t) \geq 0$; cumulative $C_t := \sum_{s=0}^{t-1} c_s$ (in wind fatigue, Miner's-rule damage). The *urgency ratio* captures the budget–time state:

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T} \tag{2}$$

where ρ_t is the remaining budget fraction and τ_t the remaining time fraction. When $u > 1$, the agent has underspent; when $u < 1$, overspent. Per-agent budgets d^i yield u_t^i computed independently per agent.

Proposition 1 (Motivating analysis under idealized Boltzmann response)**.** *Consider an idealized policy that responds to penalty weight λ via a Boltzmann model: $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, where P_0 is the base risky-action probability and $\alpha > 0$ is the policy's sensitivity. Under this model, the penalty weight that maintains budget balance has a multiplicative structure determined by the urgency ratio:*

$$w^*(u) = \frac{1}{u} \tag{3}$$

Proof sketch. Budget balance requires $(T - t) \cdot P(\text{risky} \mid \lambda) = d - C_t$. Substituting the Boltzmann model, solving for λ, and computing the multiplicative weight relative to the TWAP baseline ($u = 1$) yields $w = \tau/\rho = 1/u$. The full derivation is given in Appendix A.1. □

Practical schedule. Since $1/u \to \infty$ as $u \to 0$ (budget depleted) and the Boltzmann response is an idealization that does not hold for real policy architectures, we use a parametric family that preserves the key structural properties:

$$w(u) = \exp\left(\eta \cdot \left(\frac{1}{u} - 1\right)\right) \tag{4}$$

The parameter $\eta \geq 0$ controls risk aversion: $\eta = 0$ recovers the constant-penalty baseline (no adaptation to budget state); $\eta > 0$ increases the penalty when overspent ($u < 1$) and relaxes it when underspent ($u > 1$).

Hard wall backstop. A steep exponential barrier activates when budget drops below 5%:

$$\lambda_{\text{wall}}(t) = \exp\left(k \cdot \text{ReLU}\left(1 - \frac{\rho_t}{0.05}\right)\right) \tag{5}$$

The combined weight is $\lambda(t) = \min(w(u_t) \cdot \lambda_{\text{wall}}(t),\ \lambda_{\max})$, clamped at $\lambda_{\max} = 10^6$.

Approximation quality. The practical schedule $\exp(\eta(1/u - 1))$ matches $u^{-\eta}$ within 5% for $u \in [0.7, 1.3]$; the approximation quality is detailed in Appendix Figure 1. Ablation results (Table 5) show that the smooth schedule alone is sufficient in practice.

Penalty landscape. The TWAP diagonal ($\rho = \tau$, where $u = 1$) separates surplus ($\lambda < 1$) from deficit ($\lambda > 1$); higher η produces steeper gradients near this boundary (Appendix Figure 2).

Note on the Boltzmann assumption. The Boltzmann response is a motivating idealization; the derivation identifies u as the natural state variable and constrains the schedule to be monotone decreasing in u, but the specific form $\exp(\eta(1/u - 1))$ is a principled parametric choice rather than an optimality claim.

4 Method

The method has three components that compose independently: (1) the *budget schedule* $\lambda(t)$ from Section 3, which determines *how strictly* to enforce the constraint at each moment; (2) the *cost function* $c(s, a)$, which defines *what* is costly; and (3) the *application mechanism*, which determines *how* the penalty modifies actions. Any cost function composes with any mechanism and the same $\lambda(t)$ schedule.

4.1 Cost Functions

The simplest cost is a binary indicator $c(s, a) = \mathbf{1}[\text{yaw} < 0]$, ignoring severity. A nonlinear surrogate captures the physics: $c(s, a) = (|a|/\bar{a})^p \cdot (v(s)/v_{\text{ref}})^q$, where $p{=}2$ (quadratic yaw-fatigue) and $q{=}3$ (cubic wind-speed amplification) are physically motivated [van Dijk et al., 2025]. All cost functions compose with the same $\lambda(t)$ schedule; swapping the cost requires no retraining.

4.2 Application Mechanisms

Energy composition (EBT actors). The cost composes directly with the policy energy before action optimization: $E_{\text{total}} = \frac{1}{N}\sum_i [E^i_\phi + \lambda_{\text{gs}}\lambda^i(t)P(c(s, a^i))]$, where $P(x) = \exp(k \cdot \text{ReLU}(x)) - 1$.

Gradient correction (any actor). For standard actors, we correct the output via:

$$a_{\text{corrected}} = a_{\text{actor}} - \lambda_{\text{gs}} \cdot \lambda(t) \cdot \alpha \cdot \nabla_a\, c(s, a_{\text{actor}}) \tag{6}$$

iterated for K steps. The gradient handles nonlinear costs that action scaling cannot: $\nabla_a c$ points away from costly regions, not just toward lower magnitude.

Algorithm 1 Post-Hoc Budget-Constrained Inference

Require: Trained actor, cost limit d^i per agent, horizon T, risk aversion η, cost $c(s, a)$

$C_0^i \leftarrow 0$ for all agents i
for $t = 0, \ldots, T-1$ **do**
  Observe s_t; compute $u_t^i = \frac{(d^i - C_t)/d^i}{(T-t)/T}$, $\lambda^i(t) = \min(\exp(\eta(1/u_t^i - 1)) \cdot \lambda_{\text{wall}}^i,\ \lambda_{\max})$
  Generate a_t: {choose one mechanism}
    EBT: gradient descent on $E_\phi + \lambda_{\text{gs}} \cdot \lambda^i(t) \cdot P(c(s_t, a))$
    MLP: $a \leftarrow a_{\text{actor}} - \lambda_{\text{gs}} \cdot \lambda^i(t) \cdot \alpha \cdot \nabla_a c(s_t, a)$ for K steps
  Execute a_t, observe r_t; $c_t^i = c^i(s_t, a_t)$
  $C_{t+1}^i \leftarrow C_t + c_t^i$
end for

4.3 Algorithm

The λ computation adds 4.7 ms overhead (19.8%) on the wind farm EBT actor. The parameters $\eta \in [2, 5]$ and $\lambda_{\text{gs}} \in [0.05, 0.5]$ require mild calibration via evaluation episodes—cheaper than retraining but not zero-cost. Unlike the Lagrangian dual μ in CMDPs (learned via gradient ascent, requiring retraining), $\lambda^i(t)$ is computed in closed form, enabling instant adaptation to new budget specifications.

5 Experimental Setup

We evaluate the budget scheduling mechanism across two domains with different risky-action semantics.

5.1 Domain 1: Wind Farm Yaw Control

Environment. WindGym [Nilsen et al., 2025] with PyWake backend. We test on two layouts: a 3-turbine row and a 9-turbine 3×3 grid. Wind speed sampled from 10–14 m/s, wind direction from 225–315°, providing natural temporal variation in wake conditions.

Risky action. Negative yaw angles ($a^i < 0$) steer the rotor against the conventional direction, deflecting wakes to increase downstream power but imposing structural loads on the yawed turbine [van Dijk et al., 2025].

Budget. B = total timesteps at negative yaw per turbine per episode.

Agent. Energy-Based Transformer (EBT) actor with Soft Actor-Critic (SAC), trained unconstrained for 100k steps. The budget surrogate composes post-hoc via per-turbine energy addition at inference.

5.2 Domain 2: Safety Gymnasium Navigation

SafetyPointGoal1-v0 [Ji et al., 2023]: a point robot navigates to goals while avoiding hazard zones (c=1 on entry). Unlike the wind farm, cost is *spatial*—it depends on position, not action direction. SAC with MLP actor, trained unconstrained for 200k steps. This domain serves as a diagnostic for state-dependent costs where the action-to-cost coupling is weak.

5.3 Baselines and Ablations

Baselines: unconstrained (upper bound), hard clip (lower bound), constant penalty (η=0, static Lagrangian), and our AC schedule ($\eta \in \{2, 5\}$). Additionally, we compare against an oracle-tuned constant penalty found

Table 1: Wind farm comparison (3-turbine, budget=15/200 steps, 50 episodes). The AC schedule outperforms the oracle-tuned constant penalty (found via 15-step bisection) while requiring zero per-budget calibration.

Method	Power (MW)	% Uncon	NegYaw	Notes
Unconstrained	27.1 ± 0.6	100%	[49, 56, 61]	—
Oracle const ($k=5.86$)	26.2 ± 0.7	96.7%	[11, 13, 14]	15-step bisection
AC ($\eta=2$, gs=0.05)	26.8 ± 0.6	**98.6%**	[9, 10, 10]	Zero calibration
AC ($\eta=5$, gs=0.1)	26.9 ± 0.5	**98.9%**	[8, 9, 10]	Zero calibration

Table 2: Binary vs. nonlinear fatigue cost (3-turbine wind farm). Nonlinear: $|y/\bar{y}|^2 \cdot (v/v_{\text{ref}})^3$.

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	[66, 75, 71]	20.3
Binary $\eta=2$ (AC)	28.1	[8, 8, 10]	86.2
Nonlinear $\eta=5$ (AC)	**29.2**	[80, 81, 83]	**12.8**

via 15-step bisection search. Ablations on the wind farm test the hard guard, schedule form ($1/u^\eta$ vs. exp), and budget flexibility (Appendix).

6 Results

6.1 Wind Farm: AC Schedule vs. Oracle Baseline

Table 1 compares the AC schedule against an oracle-tuned constant penalty (15-step bisection) on the 3-turbine wind farm (50 episodes, budget=15). The AC schedule achieves 98.9% power retention with neg-yaw $[8, 9, 10]$, outperforming the oracle constant (96.7%, $[11, 13, 14]$) without any per-budget calibration.

Power retention is within 5% of unconstrained across all budget levels; the AC schedule keeps all turbines within budget while the constant penalty requires per-budget re-tuning (Appendix Figures 6–7).

6.2 Nonlinear Fatigue: Why the Cost Function Matters

Table 2 compares binary vs. nonlinear fatigue costs. The binary AC ($\eta=2$) avoids negative yaw entirely but accumulates fatigue of 86.2—*worse* than unconstrained (20.3)—because it pushes the agent toward large positive yaw angles. The nonlinear AC ($\eta=5$) uses *more* neg-yaw steps (80–83) but achieves fatigue of only 12.8 (6.8× reduction, 98% power) by permitting mild yaw at low wind speeds and restricting it at high speeds.

6.3 Ablations

The AC schedule alone keeps all turbines within budget ($[11, 11, 11]$ at $d=15$) without the hard guard; the constant penalty overshoots at $[29, 51, 58]$. The $1/u^\eta$ and $\exp(\eta(1/u-1))$ schedules perform comparably. A single policy serves budgets from 2.5% to 50% with >95% power retention from $d=10$ onward. Full ablation tables are in the Appendix.

6.4 Scope: Action-Dependent vs. State-Dependent Costs

The method succeeds when cost depends on the current action ($\nabla_a c \neq 0$). Realistic fatigue involves *state-dependent* costs—wake propagation delays decouple upstream yaw from downstream loading [van Dijk et al.,

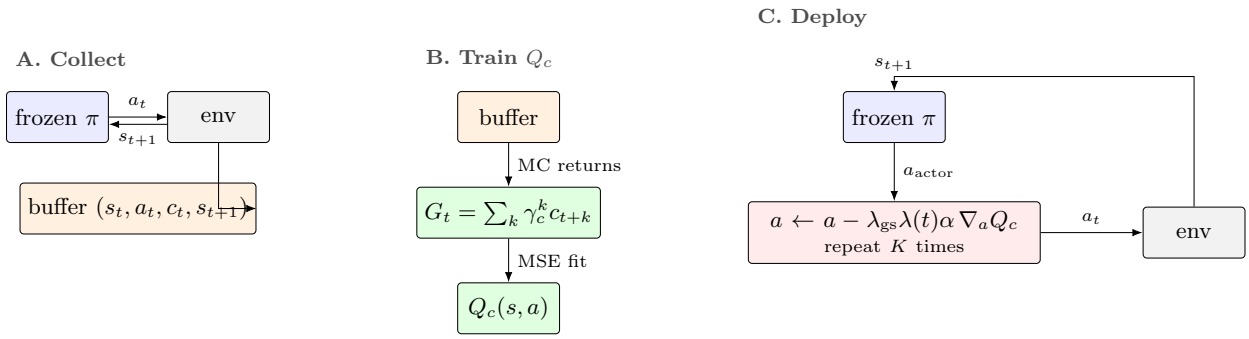

Figure 1: Q_c pipeline. **A.** Collect transitions from frozen policy rollouts. **B.** Fit Q_c on Monte Carlo cost returns (no bootstrap). **C.** Deploy: actor proposes action, Q_c gradient corrects it K times per step. Policy stays frozen throughout.

2025]. Safety Gym [Ji et al., 2023] exhibits the same structure: position-dependent cost with momentum dynamics.

To bridge this gap, we train a cost critic $Q_c(s, a) = \mathbb{E}_\pi[\sum \gamma_c^k c_{t+k}]$ offline (MC returns, $\gamma_c = 0.99$, correlation 0.965) and modulate $\nabla_a Q_c$ via the urgency schedule (Figure **??**).

On the wind farm (strong coupling), Q_c achieves 5.7× fatigue reduction; on Safety Gym (weak coupling), 20–36% cost reduction at >97% reward, satisfying budget at $d = 40$ but not tighter (Appendix Tables 4, 3). The *coupling strength*—$\|\nabla_a Q_c\|$ relative to $\|\nabla_s Q_c\|$—determines the correction ceiling.

7 Related Work

Constrained MDPs. Lagrangian CMDP methods [Altman, 1999, **?**, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with remaining budget ρ_t (same variable we use) but still retrains the policy, so each new budget requires a new policy. TREBI [Lin et al., 2023] handles real-time budget changes via diffusion models but requires offline data and diffusion-specific training. Our method instead freezes the policy and uses a closed-form $\lambda(t)$ schedule at inference—no retraining for any budget.

Post-hoc constraint composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], control barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle *instantaneous* constraints; none address cumulative budgets spanning an episode.

Budget pacing and optimal execution. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] provide pacing strategies. Our work adopts the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001]—remaining inventory vs. remaining time—without claiming structural equivalence to market-impact models.

8 Conclusion

We introduced a closed-form urgency-ratio schedule for enforcing cumulative budget constraints on pre-trained RL policies without retraining. On wind farm yaw control, a nonlinear fatigue surrogate achieves 6.8× fatigue reduction while retaining 98% of power; a learned cost critic Q_c discovers the same physics from data (5.7× reduction). On Safety Gym, Q_c achieves 20–36% cost reduction at >97% reward, with the action-to-cost coupling strength determining the ceiling. The method requires zero policy retraining but mild calibration of η and λ_{gs}; the Q_c extension requires offline training on logged transitions. For state-dependent costs where action-to-cost coupling is weak (momentum-dominated dynamics, wake propagation delays), future work on

control barrier functions or model-predictive shielding may close the gap. Code and checkpoints are available at [anonymized].

A Additional Results and Proofs

A.1 Full Proof of Proposition 1

Proof. Budget balance requires that the expected number of risky actions over the remaining horizon equals the remaining budget:

$$(T - t) \cdot P(\text{risky} \mid \lambda) = d - C_t \tag{7}$$

Substituting the Boltzmann model: $(T - t) \cdot P_0 \exp(-\alpha\lambda) = d - C_t$. Solving: $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)\, P_0}{d - C_t}$. At the TWAP baseline ($u = 1$, where $\rho = \tau$, *i.e.*, $(d - C_t)/d = (T - t)/T$), the budget balance gives $\lambda_0 = \frac{1}{\alpha} \ln \frac{T P_0}{d}$. The multiplicative weight relative to the TWAP penalty is:

$$w = \exp\big(\alpha(\lambda - \lambda_0)\big) = \frac{(T - t)\, P_0}{d - C_t} \cdot \frac{d}{T\, P_0} = \frac{(T - t)/T}{(d - C_t)/d} = \frac{\tau}{\rho} = \frac{1}{u}$$

□

A.2 Approximation Quality

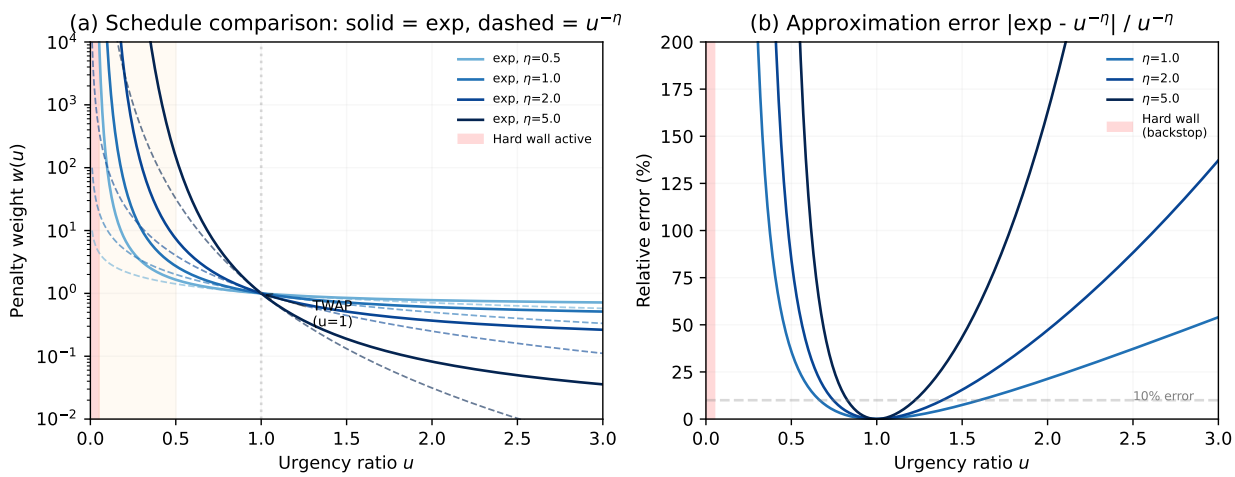

Figure 2: (a) Schedule comparison: $\exp(\eta(1/u - 1))$ (solid) vs. $u^{-\eta}$ (dashed) across the full urgency range. Red region: hard wall backstop active ($u < 0.05$). (b) Relative approximation error. The schedules agree within 10% for $u > 0.5$ (the regime where smooth pacing operates).

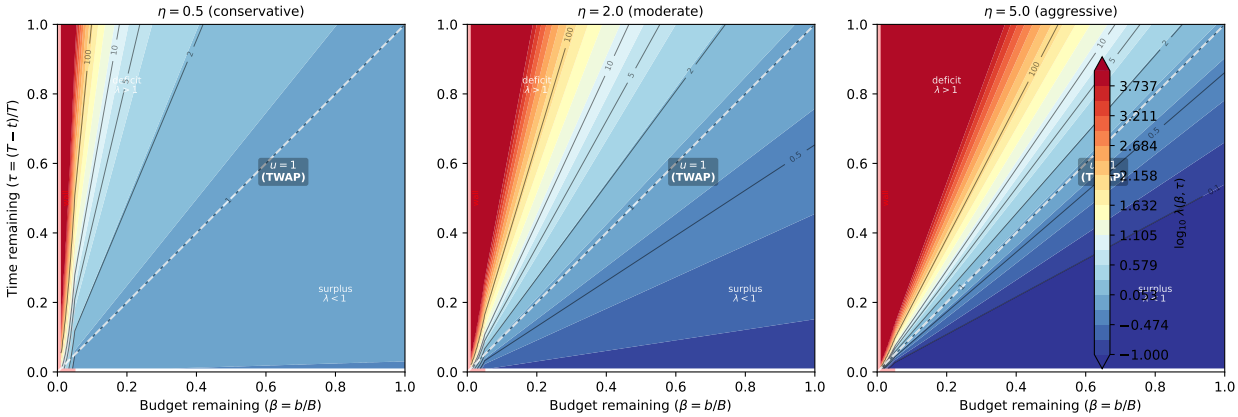

Figure 3: Penalty weight $\lambda(\rho, \tau)$ across the budget–time state space for three risk-aversion values η. The TWAP diagonal (white dashed, $u = \rho/\tau = 1$) separates surplus (blue, $\lambda < 1$) from deficit (red, $\lambda > 1$). Contour lines show penalty magnitudes. Higher η produces steeper gradients near the diagonal, concentrating budget spending into favorable conditions.

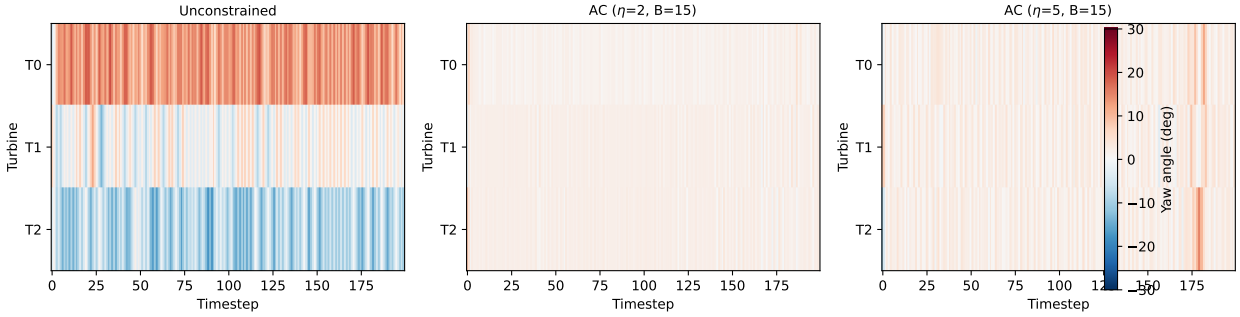

Figure 4: Yaw angle heatmaps (turbine × timestep) showing the spatial-temporal pattern. Unconstrained (left): persistent positive/negative yaw steering across all turbines. AC $\eta = 2$ (middle): nearly zero yaw—very conservative. AC $\eta = 5$ (right): mostly zero with a concentrated burst of negative yaw (blue) near the end, representing strategic budget spending.

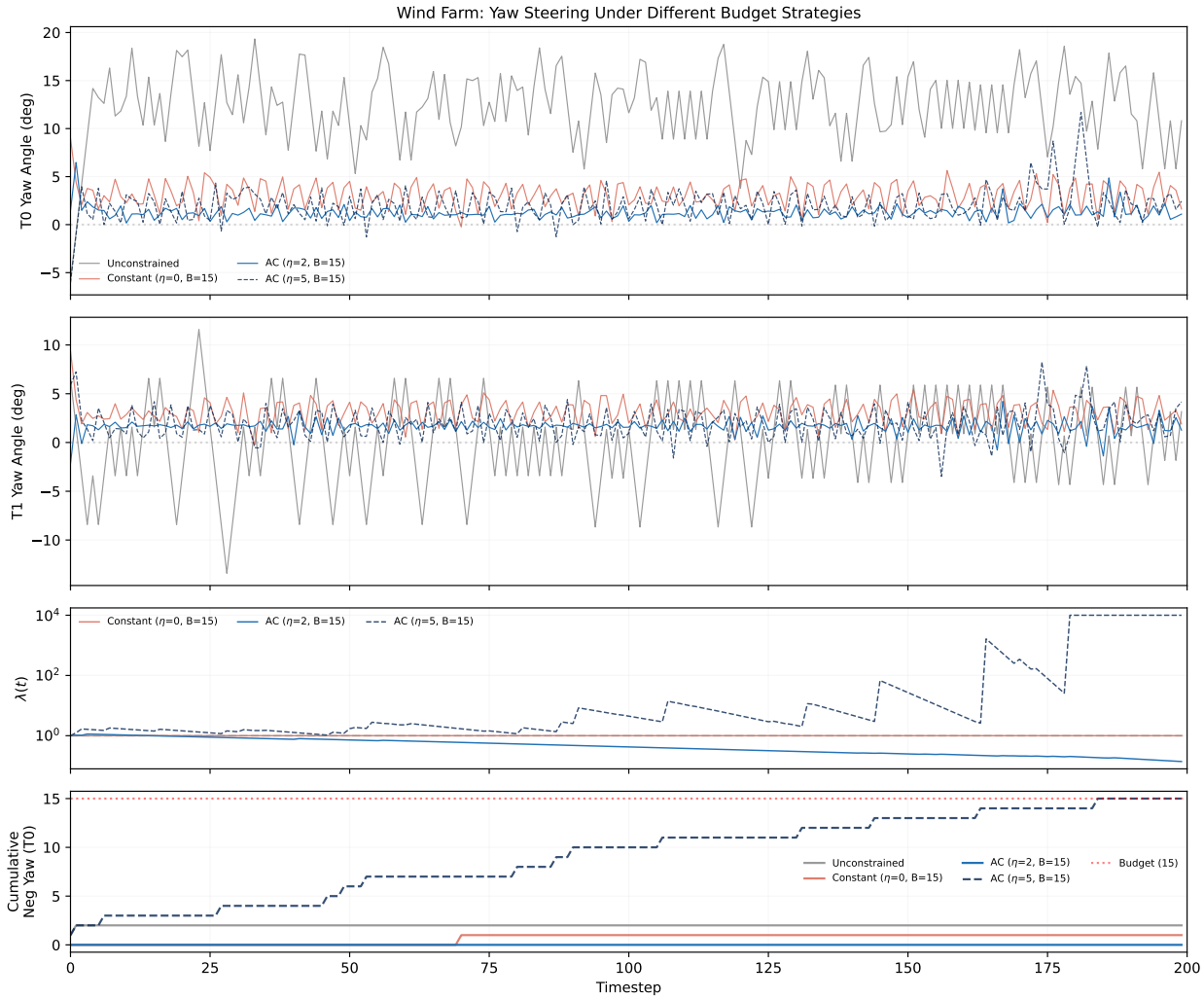

Figure 5: Wind farm yaw time series (one episode, budget= 15). $\lambda(t)$ adapts: $\eta = 5$ drops during surplus and spikes near depletion. Cumulative neg-yaw shows $\eta = 5$ uses the full budget.

A.3 Penalty Landscape

A.4 Yaw Angle Heatmaps

A.5 Wind Farm Time Series and Pareto

A.6 Safety Gymnasium Diagnostic

A.7 Learned Cost Critic on Wind Farm

A.8 Wind Farm Spatial Allocation

A.9 Hard Guard Ablation

A.10 Schedule Comparison: $1/u^\eta$ vs. $\exp(\eta(1/u - 1))$

A.11 Budget Flexibility

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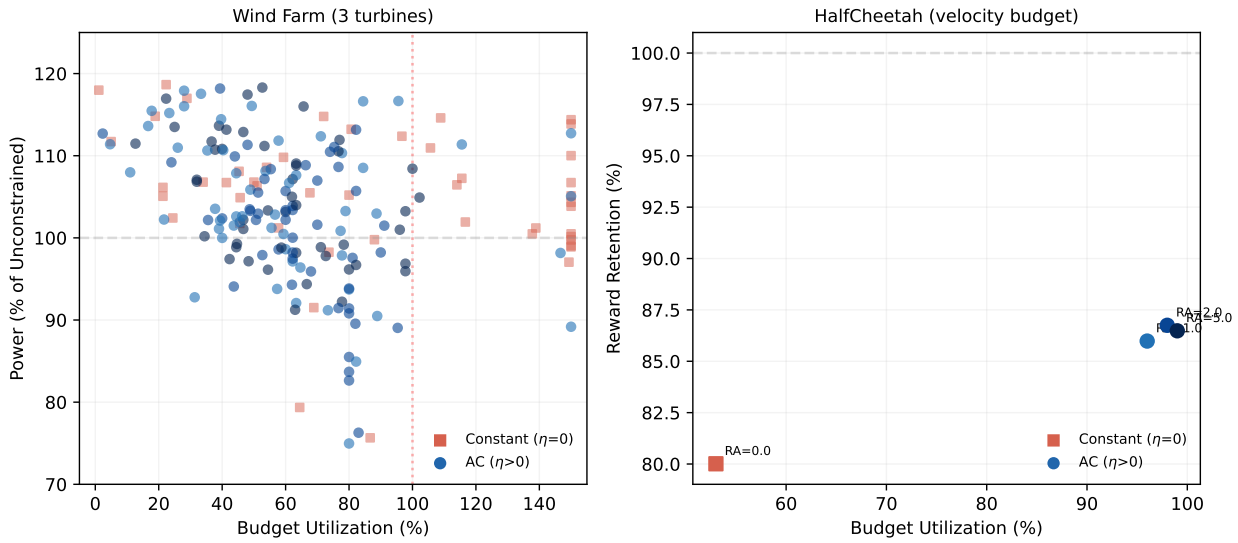

Figure 6: Budget utilization vs. reward retention. Constant penalty (red) cannot achieve both; AC schedule (blue) clusters in the efficient top-right region.

Table 3: Safety Gymnasium diagnostic (SafetyPointGoal1-v0, 20 episodes). Q_c reduces cost 20–36% while preserving >97% reward. Budget satisfied only at $d=40$.

Method	Budget	Reward	% Uncon	Cost	OK
Unconstrained	—	26.7 ± 1.1	100%	∼50	—
Q_c gradient correction, $\eta = 5$ (fixed config)					
Q_c grad	10	26.1	98%	41	×
Q_c grad	25	26.7	100%	41	×
Q_c grad	40	26.6	100%	55	×
Best across modes (ceiling of post-hoc correction)					
Q_c sample ($\eta=5$)	10	26.2	98%	40	×
Q_c sample ($\gamma_c=0.95$)	25	26.7	99%	**29**	×
Q_c grad ($\eta=0$)	40	25.8	97%	**33**	✓

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Table 4: Learned cost critic Q_c on 3-turbine wind farm (10 episodes). Q_c trained on 100 episodes with nonlinear fatigue cost. Power exceeding unconstrained reflects Q_c steering toward configurations that are both lower-fatigue and higher-power.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6 ± 5.6	[84, 96, 94]	54.6	1.0×
Q_c, $\eta = 2$, gs=0.05	$\mathbf{28.1} \pm 2.7$	[43, 44, 43]	**9.5**	5.7×
Q_c, $\eta = 5$, gs=0.5	28.8 ± 2.0	[55, 54, 51]	10.4	5.3×
Analytical surr. $\eta = 5$	29.2	[80, 81, 83]	12.8	4.3×

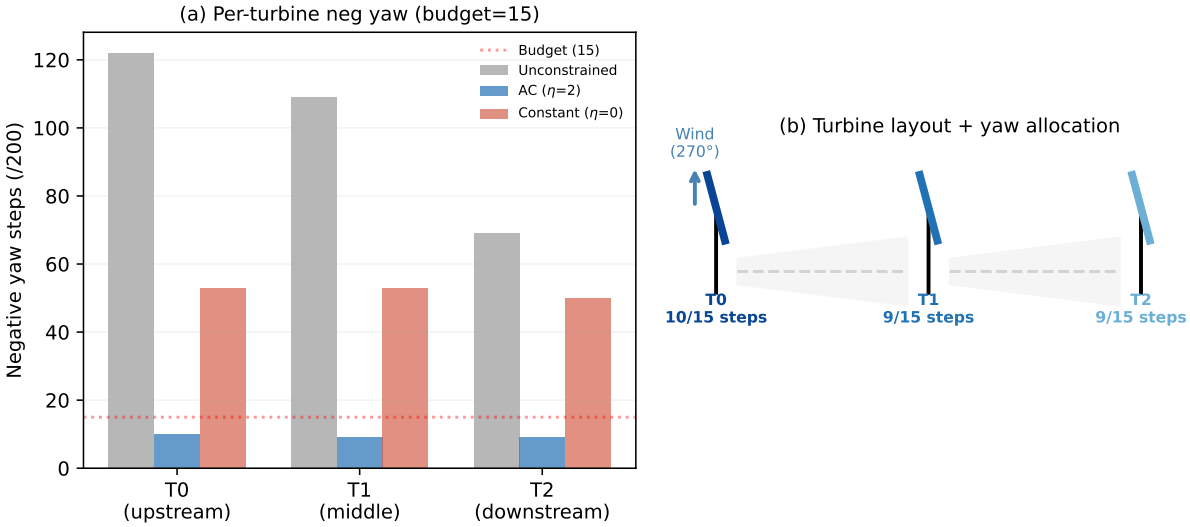

Figure 7: Wind farm spatial yaw allocation. (a) Unconstrained agent uses negative yaw heavily on all turbines; AC ($\eta = 2$) constrains to within the 15-step budget; constant penalty over-spends. (b) Turbine layout.

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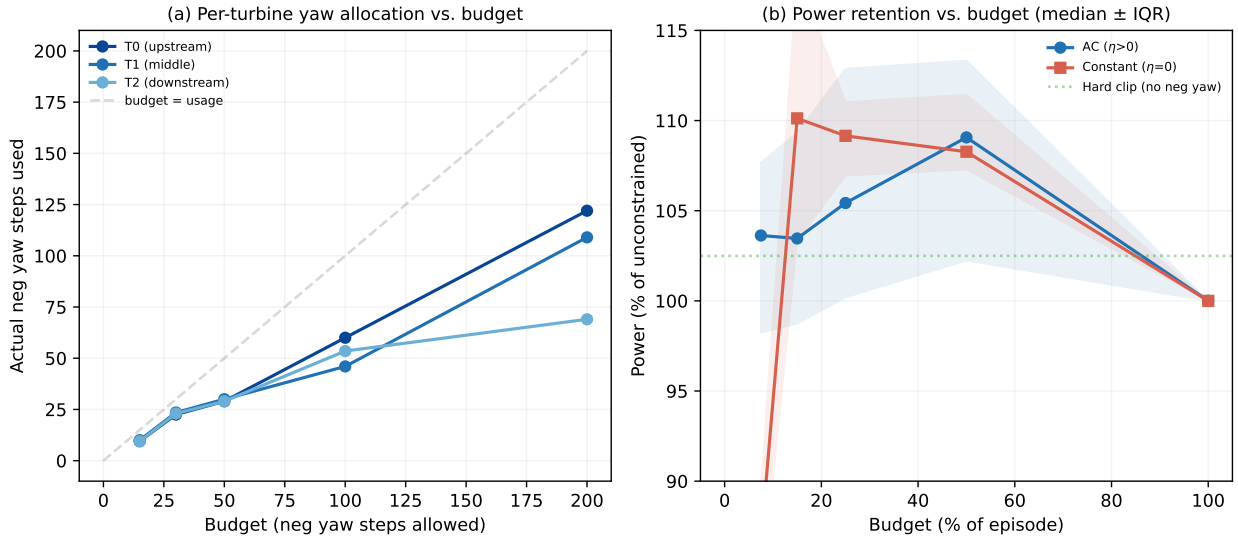

Figure 8: Wind farm budget effect. (a) Per-turbine allocation: as budget tightens, downstream T2 reduces negative yaw first. (b) Power retention remains within 5% of unconstrained.

Table 5: Hard guard ablation (3-turbine wind farm, budget=15/200 steps, $\eta = 2$, $k = 2$, gs=0.1, 10 episodes).

Configuration	Power (MW)	NegYaw [T0, T1, T2]	Budget
Unconstrained	26.8 ± 3.9	[83, 87, 100]	—
Both (AC + guard)	24.7 ± 5.1	[13, 16, 19]	✓
AC only (no guard)	26.3 ± 3.0	[11, 11, 11]	✓
Guard only ($\eta = 0$)	24.8 ± 5.7	[29, 51, 58]	×

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Table 6: Schedule comparison on 3-turbine wind farm (budget=15/200 steps, $k = 2$, gs=0.1, 10 episodes).

Schedule	η	Power (MW)	NegYaw [T0, T1, T2]
$\exp(\eta(1/u - 1))$	0.5	24.3 ± 6.3	[32, 36, 37]
$\exp(\eta(1/u - 1))$	2.0	25.5 ± 5.3	[19, 24, 24]
$\exp(\eta(1/u - 1))$	5.0	27.8 ± 4.5	[14, 14, 14]
$u^{-\eta}$	0.5	24.9 ± 5.6	[56, 57, 58]
$u^{-\eta}$	2.0	24.8 ± 6.8	[29, 34, 35]
$u^{-\eta}$	5.0	26.6 ± 5.3	[22, 24, 24]

Table 7: Budget flexibility (3-turbine wind farm, $\eta = 2$, $k = 2$, gs=0.1, 10 episodes). A single policy serves arbitrary budget levels. Power retention is high from $d = 10$ onward.

Budget (steps/200)	Power (MW)	% Unconstrained	NegYaw [T0, T1, T2]
5 (2.5%)	26.4 ± 3.9	100.1%	[74, 76, 86]
10 (5%)	25.2 ± 5.3	95.7%	[25, 25, 26]
15 (7.5%)	29.2 ± 1.8	110.9%	[20, 20, 21]
25 (12.5%)	26.1 ± 4.1	99.1%	[17, 20, 20]
50 (25%)	26.2 ± 6.2	99.3%	[27, 39, 39]
75 (37.5%)	27.7 ± 3.6	105.2%	[54, 53, 62]
100 (50%)	22.3 ± 6.1	84.9%	[33, 50, 74]